

A FIRST-JET CHARACTERIZATION OF GRIFFITHS POSITIVITY

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ABSTRACT. We characterize Griffiths positivity of vector bundles on smooth projective varieties by a finite-dimensional convex condition on first jets of sections of positive twists. A nonzero positive matrix-valued measure annihilated by the adjoint Levi operator gives the dual obstruction. We also give an explicit bundle on an abelian surface that satisfies the first-level condition, but whose complete untwisted evaluation has a differential kernel.

1. INTRODUCTION AND MAIN RESULT

Let $E \rightarrow X$ be a holomorphic vector bundle of positive rank over a connected smooth complex projective variety. Griffiths [8] proved that Griffiths positivity implies ampleness and conjectured the converse. Du–Xie [7] show that ampleness alone does not determine the existence of a Griffiths-positive metric. The question is which data of a holomorphic bundle detect Griffiths positivity.

Related recognition questions arise for cotangent bundles of complete intersections. The high-degree ampleness statement proposed by Debarre [3] was first proved in full generality by Xie [19], using explicit differential forms of Brotbek [1]. A subsequent proof was given by Brotbek–Darondeau [2]. Under a stronger codimension hypothesis, Mohsen [16] constructed complete intersections with Griffiths-positive cotangent metrics. In the local model of that construction, positivity is expressed by immersivity of a Gauss map [16, Lemma 13].

Deng–Ning–Wang–Zhou [5, Theorems 1.2 and 1.3] give criteria for Griffiths semipositivity in terms of L^p estimates for $\bar{\partial}$ and holomorphic extension conditions. For a fixed smooth Hermitian metric, Liu–Xu [15, Theorem 1.1] characterize Griffiths curvature lower bounds by local L^2 extension conditions. Demailly [4] approaches the existence of positive metrics through a nonlinear elliptic system of Hermitian–Yang–Mills type.

A complementary approach to geometric positivity comes from duality between positive forms and currents, as in Sullivan’s [17] theory of structure currents and Harvey–Lawson’s [10, Theorem 14] characterization of Kähler manifolds. We use such a duality to study the global existence of a Griffiths-positive metric, by expressing curvature positivity as a linear condition on Hermitian forms.

Ampleness controls the tautological line bundle on a projectivized bundle, while Griffiths positivity is a condition on a single Hermitian quadratic norm on each vector-space fibre; complex Finsler geometry makes this distinction explicit [13]. A second difficulty is that the Chern curvature is nonlinear in the metric coefficients. For a real C^2 function u on a complex manifold, its Levi form is the Hermitian form $\text{Levi}(u)(\xi, \eta) = (\partial\bar{\partial}u)(\xi, \bar{\eta})$ on $(1, 0)$ tangent vectors. It is strictly positive at a point p if $\text{Levi}(u)_p(\xi, \xi) > 0$ for every nonzero $(1, 0)$ tangent vector ξ at p ; u is strictly plurisubharmonic when this holds at every point. Applying this operator to fibrewise Hermitian quadratic functions on the dual bundle turns the metric existence problem into a linear positivity condition.

Write $F := E^*$ for the dual bundle. A smooth Hermitian form h on F , not assumed positive, determines the fibrewise quadratic function

$$G_h(v) := h(v, v).$$

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Throughout, $\text{Levi}(G_h)$ is taken on the total space of F , so its arguments include both base and fibre directions. It is real linear in h . For a positive-definite metric h , the condition $\text{Levi}(G_h)_v(\xi, \xi) > 0$ for every $v \in F \setminus 0$ and every nonzero $\xi \in T_v^{1,0}F$ is equivalent to Griffiths negativity of h , hence to Griffiths positivity of the dual metric on E . This linearization gives an exact theorem of alternatives: the obstruction is a nonzero positive matrix-valued measure annihilating the Levi forms of all fibrewise Hermitian quadratic functions. After fixing a projective polarization, Griffiths positivity is detected at some finite level by first jets of global sections, whereas failure at every level yields such a measure. The first jet of a holomorphic section at a point records its value and first derivatives in local holomorphic coordinates and a local holomorphic frame; two sections have the same first jet precisely when their difference vanishes to order at least two at that point.

Fix a very ample line bundle A with the Fubini–Study metric a induced by a positive Hermitian inner product on $H^0(X, A)$. Since A is ample, Serre’s global generation theorem allows us to choose an integer $m_0 \geq 0$ such that $E \otimes A^{m_0}$ is globally generated. For $m \geq m_0$ set

$$V_m := H^0(X, E \otimes A^m), \quad N_m := \dim V_m.$$

Choose a basis $s_1^{(m)}, \dots, s_{N_m}^{(m)}$ of V_m , and let $\text{Herm}(N_m)$ be the real vector space of Hermitian $N_m \times N_m$ matrices. A matrix $B \in \text{Herm}(N_m)$ determines a smooth Hermitian form $h_{m,B}$ on F by

$$G_{m,B}(x, v) = h_{m,B,x}(v, v) := \sum_{i,j=1}^{N_m} B_{ij} a_x^m(v(s_j^{(m)}(x)), v(s_i^{(m)}(x))).$$

Here $v \in F_x$, and a^m pairs the evaluated sections in A_x^m , linearly in its first argument. The forms $h_{m,B}$ comprise the real vector space $\mathcal{H}_m$; no positivity assumption is imposed on B . This is the construction of Bergman sections for arbitrary Hermitian forms in Lempert [14, Sections 3 and 4], applied to $E \otimes A^m$ and contracted with a^m .

Fix a background Hermitian metric k on F , with unit sphere bundle $S_k F := \{v \in F : k(v, v) = 1\}$, and a Hermitian metric b on $W := T^{1,0}F|_{S_k F}$. The corresponding unit tangent vectors form the compact space

$$\mathcal{K} := \{(v, \xi) : v \in S_k F, \xi \in T_v^{1,0}F, b(\xi, \xi) = 1\}. \quad (1.1)$$

For a general Hermitian form h write

$$\mathcal{D}h := \text{Levi}(G_h)|_{S_k F}.$$

A positive matrix-valued Radon measure T on $S_k F$ acts on continuous Hermitian forms on W and is nonnegative on pointwise positive semidefinite forms. The distribution $\mathcal{D}^*T$ on X acts on smooth Hermitian forms on F by

$$\langle \mathcal{D}^*T, h \rangle = \langle T, \mathcal{D}h \rangle.$$

Our main theorem characterizes Griffiths positivity both by a finite-level first-jet condition and by the absence of a positive measure obstruction.

Theorem 1.1. *For every $m \geq m_0$ there is a continuous map*

$$\Psi_m : \mathcal{K} \longrightarrow \text{Herm}(N_m)$$

from the compact space $\mathcal{K}$ in (1.1) to the finite-dimensional real vector space $\text{Herm}(N_m)$, satisfying

$$\text{Levi}(G_{m,B})_v(\xi, \xi) = \text{tr}(B\Psi_m(v, \xi)). \quad (1.2)$$

The following conditions are equivalent:

- (i) *E admits a smooth Griffiths-positive Hermitian metric;*
- (ii) *for some $m \geq m_0$ and some $B \in \text{Herm}(N_m)$,*

$$\text{tr}(B\Psi_m(q)) > 0 \quad (q \in \mathcal{K});$$

(iii) for some $m \geq m_0$,

$$0 \notin \text{conv } \Psi_m(\mathcal{K}),$$

where conv denotes the convex hull;

(iv) there is no nonzero positive matrix-valued Radon measure T with $\mathcal{D}^*T = 0$.

Level m succeeds if the inequality in (ii) holds for some B , equivalently if this trace pairing has a positive minimum on $\mathcal{K}$. Some finite level succeeds precisely when E is Griffiths positive, so this existence condition depends only on E ; otherwise every level fails. When level m fails, Proposition 2.1 gives a probability measure μ_m supported on at most $N_m^2 + 1$ points of $\mathcal{K}$ and satisfying $\int_{\mathcal{K}} \Psi_m d\mu_m = 0$.

The proof begins by expressing the Levi form at each level through first jets of global sections of $E \otimes A^m$ and of the polarization A . Allowing arbitrary Hermitian coefficient matrices makes $\mathcal{H}_m$ a real linear space, so finite-dimensional separation gives either a strictly positive trace pairing or an atomic probability measure annihilating that level.

The passage from finite levels to all smooth metrics rests on nestedness and C^2 -density. The Fubini–Study identity makes the spaces $\mathcal{H}_m$ nested, while the Bergman kernel expansion in Wang [18, Appendix, Theorems 5.1 and 5.2] supplies the approximation. For a Griffiths-negative metric h on F , the inequality $\mathcal{D}h > 0$ on the compact sphere bundle $S_k F$ persists under sufficiently close C^2 approximation, giving success at a finite level. If every level fails, a weak limit of the atomic probability measures annihilates all finite levels by nestedness, and then all smooth Hermitian forms by density. This limit induces a nonzero positive matrix-valued measure T satisfying $\mathcal{D}^*T = 0$.

The first-jet identity (1.2) gives a geometric interpretation of the untwisted level: for a globally generated bundle E , level zero succeeds precisely when the complete evaluation map

$$E^* \setminus 0 \longrightarrow H^0(X, E)^*$$

is an immersion. This is close to the Gauss-map criterion in Mohsen’s complete-intersection construction [16, Lemma 13]. Twisting can recover tangent directions lost by the untwisted evaluation map, as the example below shows.

We apply the criterion to an explicit pullback $E := f^*S$ of a Steiner bundle, of the type used as a comparison bundle in Du–Xie [7]. This pullback is different from the counterexample in that paper: the counterexample was obtained later by deformation and a moduli-space argument, whereas the present pullback is Griffiths positive. In Section 3 we show that ramification obstructs level zero and construct a first-level matrix satisfying the first-jet inequality of Theorem 1.1. This gives an explicit case in which twisting detects positivity despite the differential kernel of the complete untwisted evaluation.

Example 1.2. For the elliptic curve $C : y^2 = x^3 - x$ with point at infinity ∞ , consider

$$X := C \times C, \quad H := \mathcal{O}_C(4[\infty]) \boxtimes \mathcal{O}_C(5[\infty]).$$

Here $L_1 \boxtimes L_2 = \text{pr}_1^* L_1 \otimes \text{pr}_2^* L_2$ denotes the external tensor product, where $\text{pr}_i : X \rightarrow C$ are the two projections. There is an explicit rank-two bundle E on X with the following properties for the polarization H endowed with a suitable Fubini–Study metric:

- (i) The complete level-zero evaluation has a nonzero differential kernel, so level zero fails.
- (ii) At level one the Hermitian matrix B_1 in (3.7) satisfies

$$\text{tr}(B_1 \Psi_1(q)) > 0 \quad (q \in \mathcal{K}),$$

so level one succeeds.

Theorem 1.1 applied to (ii) gives Griffiths positivity of E . Together with (i), this shows how twisting by H detects positivity despite the differential kernel of the complete untwisted evaluation.

Plan of the paper. Section 2 develops the Levi operator and proves Theorem 1.1 through the first-jet formula, the finite-dimensional alternative, and C^2 -density. In Section 3, we

verify Example 1.2: first the failure at level zero, then the construction of a first-level matrix and verification of the strict inequality.

2. FINITE-DIMENSIONAL REDUCTION OF GRIFFITHS POSITIVITY

For a real C^2 function u , we write $\text{Levi}(u)(\xi, \eta) = (\partial\bar{\partial}u)(\xi, \bar{\eta})$ for its Levi form on $(1, 0)$ tangent vectors. Let h be a smooth Hermitian form on $F = E^*$, not assumed positive definite, with fibrewise quadratic function $G_h(v) = h(v, v)$. At a point $v \in F_x \setminus 0$, a vector $w \in F_x$ represents a vertical tangent vector through the holomorphic curve $t \mapsto v + tw$. Restricting G_h to this curve gives

$$\text{Levi}(G_h)_v(w, w) = \left. \frac{\partial^2}{\partial t \partial \bar{t}} h_x(v + tw, v + tw) \right|_{t=0} = h_x(w, w). \quad (2.1)$$

If G_h is strictly plurisubharmonic on $F \setminus 0$, the left-hand side is positive for every $w \neq 0$. Since x is arbitrary, h is then a positive-definite metric on F . For a smooth positive-definite Hermitian metric h on F , Griffiths negativity of h is equivalent to strict plurisubharmonicity of G_h on $F \setminus 0$, by Drinovec Drnovšek–Forstnerič [6, Proposition 6.2(ii)–(iii)].

In a local holomorphic frame e_A of A , write $|e_A|_a^2 = e^{-\phi}$ for the fixed metric a . Write $V_m = H^0(X, E \otimes A^m)$ and $N_m = \dim V_m$, and choose a basis $s_1^{(m)}, \dots, s_{N_m}^{(m)}$ of V_m . Vectors are written as columns. For $v \in F_x$, let $p_m(x, v) \in \mathbb{C}^{N_m}$ be the column determined by

$$v(s_j^{(m)}(x)) = (p_m(x, v))_j e_A(x)^m, \quad 1 \leq j \leq N_m. \quad (2.2)$$

The map p_m is holomorphic on the total space of F over the trivializing open set. Let $\text{Herm}(N_m)$ be the real vector space of Hermitian $N_m \times N_m$ matrices, and for $B \in \text{Herm}(N_m)$ let $h_{m,B}$ denote the smooth Hermitian form on F with local expression

$$G_{m,B}(x, v) = h_{m,B,x}(v, v) = e^{-m\phi(x)} p_m(x, v)^\dagger B p_m(x, v). \quad (2.3)$$

Here $\dagger$ denotes conjugate transpose. Under a change of local frame $e'_A = \gamma e_A$, with γ holomorphic and nowhere vanishing,

$$p'_m = \gamma^{-m} p_m, \quad \phi' = \phi - \log |\gamma|^2,$$

so (2.3) gives the same Hermitian form on overlaps. A change of basis of V_m replaces p_m by Cp_m for a constant invertible matrix C , and

$$(Cp_m)^\dagger B (Cp_m) = p_m^\dagger (C^\dagger B C) p_m.$$

Since $B \mapsto C^\dagger B C$ is a bijection of $\text{Herm}(N_m)$, the space $\mathcal{H}_m = \{h_{m,B} : B \in \text{Herm}(N_m)\}$ is independent of the chosen basis.

For $q = (v, \xi) \in \mathcal{K}$, write $\xi = (u, w)$ in a local holomorphic trivialization, with base component u and fibre component w . For $u = \sum_i u_i \partial / \partial z_i$, the derivatives of the local weight are $\phi_u = \sum_i u_i \partial_i \phi$ and $\phi_{u\bar{u}} = \sum_{i,j} u_i \bar{u}_j \partial_i \partial_{\bar{j}} \phi$. Set

$$r_m := dp_m(\xi) - m\phi_u p_m, \quad \kappa := \phi_{u\bar{u}}.$$

Differentiating (2.3) gives

$$\begin{aligned} \text{Levi}(G_{m,B})(\xi, \xi) &= e^{-m\phi} (dp_m(\xi)^\dagger B dp_m(\xi) - 2m \text{Re}(\phi_u dp_m(\xi)^\dagger B p_m) \\ &\quad + (m^2 |\phi_u|^2 - m\kappa) p_m^\dagger B p_m) \\ &= e^{-m\phi} (r_m^\dagger B r_m - m\kappa p_m^\dagger B p_m). \end{aligned}$$

This calculation leads to the local matrix-valued map

$$\Psi_m := e^{-m\phi} (r_m r_m^\dagger - m\kappa p_m p_m^\dagger), \quad (2.4)$$

for which $\text{Levi}(G_{m,B})_v(\xi, \xi) = \text{tr}(B \Psi_m(v, \xi))$ for every Hermitian matrix B . Under the frame change above, $r'_m = \gamma^{-m} r_m$ and $\kappa' = \kappa$, so the expression in (2.4) is unchanged. Thus, for the chosen basis of V_m , these local expressions yield a continuous global map $\Psi_m : \mathcal{K} \rightarrow \text{Herm}(N_m)$ satisfying (1.2), as required in Theorem 1.1.

To express ϕ_u and κ in (2.4) through first jets, choose an orthonormal basis $t_0, \dots, t_N$ for the fixed inner product on $H^0(X, A)$. Since a is the induced Fubini–Study metric,

$$e^\phi = \sum_{\alpha=0}^N |t_\alpha|^2, \quad (2.5)$$

where t_α also denotes the coefficient of the section in the frame e_A . Using superscript t for transpose, write $t := (t_0, \dots, t_N)^t$ and $\rho := t^\dagger t$. Differentiating $\phi = \log \rho$ gives

$$\phi_u = \frac{t^\dagger dt(u)}{\rho}, \quad \kappa = \phi_{u\bar{u}} = \frac{\rho |dt(u)|^2 - |t^\dagger dt(u)|^2}{\rho^2}.$$

Thus (2.4) depends only on first jets of sections of $E \otimes A^m$ and of the fixed polarization A .

Proposition 2.1. *For a fixed $m \geq m_0$, a Hermitian matrix B with $\text{tr}(B\Psi_m) > 0$ on $\mathcal{K}$ exists if and only if $0 \notin \text{conv } \Psi_m(\mathcal{K})$. If $0 \in \text{conv } \Psi_m(\mathcal{K})$, there are points $q_1, \dots, q_s \in \mathcal{K}$ and weights $\lambda_j \geq 0$ such that*

$$s \leq N_m^2 + 1, \quad \sum_{j=1}^s \lambda_j = 1, \quad \sum_{j=1}^s \lambda_j \Psi_m(q_j) = 0. \quad (2.6)$$

Proof. Let $C_m := \text{conv } \Psi_m(\mathcal{K})$. Carathéodory’s theorem, applied to $\Psi_m(\mathcal{K})$ in the real vector space $\text{Herm}(N_m)$ of dimension N_m^2 , shows that every $Z \in C_m$ can be written as

$$Z = \sum_{j=1}^{N_m^2+1} \lambda_j \Psi_m(q_j), \quad q_j \in \mathcal{K}, \quad \lambda_j \geq 0, \quad \sum_{j=1}^{N_m^2+1} \lambda_j = 1.$$

Thus the continuous map $((q_j), (\lambda_j)) \mapsto \sum_j \lambda_j \Psi_m(q_j)$ maps the compact space $\mathcal{K}^{N_m^2+1}$ times the closed weight simplex onto C_m , proving that C_m is compact. If $0 \in C_m$, taking $Z = 0$ in this representation and discarding zero weights gives (2.6). No Hermitian B can then satisfy $\text{tr}(B\Psi_m(q)) > 0$ for every $q \in \mathcal{K}$, since the weighted sum of these traces is zero. If $0 \notin C_m$, the strict separation theorem gives a real-linear functional ℓ on $\text{Herm}(N_m)$ and a constant $c > 0$ such that $\ell(Z) \geq c$ for every $Z \in C_m$, while $\ell(0) = 0$. The pairing $(B, Z) \mapsto \text{tr}(BZ)$ is a real inner product on $\text{Herm}(N_m)$. Choose an orthonormal basis $E_1, \dots, E_{N_m^2}$ for this inner product and set

$$B := \sum_{a=1}^{N_m^2} \ell(E_a) E_a \in \text{Herm}(N_m).$$

For $Z = \sum_a z_a E_a$ with $z_a \in \mathbb{R}$, orthonormality and real linearity give

$$\text{tr}(BZ) = \sum_a \ell(E_a) z_a = \ell\left(\sum_a z_a E_a\right) = \ell(Z).$$

Hence $\text{tr}(BZ) \geq c > 0$ for every $Z \in C_m$. $\square$

On the compact space $\mathcal{K} = \{(v, \xi) : v \in S_k F, \xi \in T_v^{1,0} F, b(\xi, \xi) = 1\}$ of unit tangent vectors over $S_k F = \{v \in F : k(v, v) = 1\}$, with the fixed Hermitian metrics k on F and b on $W = T^{1,0} F|_{S_k F}$, set

$$f_h(v, \xi) := \text{Levi}(G_h)_v(\xi, \xi).$$

Let $m \geq m_0$, where m_0 is the integer chosen so that $E \otimes A^{m_0}$ is globally generated. Our choice of the Fubini–Study metric gives $\sum_{\alpha=0}^N e^{-\phi} |t_\alpha|^2 = 1$ by (2.5), hence

$$G_{m,B} = e^{-(m+1)\phi} \sum_{\alpha=0}^N (p_m t_\alpha)^\dagger B (p_m t_\alpha).$$

Express the product sections in the chosen basis of V_{m+1} :

$$s_j^{(m)} t_\alpha = \sum_{\ell=1}^{N_{m+1}} (C_\alpha)_{j\ell} s_\ell^{(m+1)}, \quad C_\alpha \in \text{Mat}_{N_m \times N_{m+1}}(\mathbb{C}).$$

By (2.2), applying $v \in F_x = E_x^*$ to the E_x components of this identity and taking coefficients in e_A^{m+1} gives $p_m(x, v)t_\alpha(x) = C_\alpha p_{m+1}(x, v)$, where $t_\alpha(x)$ denotes the local coefficient of the section. Substitution into the preceding identity yields

$$G_{m,B} = e^{-(m+1)\phi} p_{m+1}^\dagger \left(\sum_{\alpha=0}^N C_\alpha^\dagger B C_\alpha \right) p_{m+1} = G_{m+1,B'}, \quad B' := \sum_{\alpha=0}^N C_\alpha^\dagger B C_\alpha.$$

Since B' is Hermitian, $h_{m,B} = h_{m+1,B'} \in \mathcal{H}_{m+1}$ for every $B \in \text{Herm}(N_m)$. Therefore $\mathcal{H}_m \subset \mathcal{H}_{m+1}$ for every $m \geq m_0$.

We will use approximation of metrics in the C^2 topology. In a local holomorphic frame, let H_h denote the coefficient matrix of h , so that $h(v, w) = w^\dagger H_h v$ for the coordinate columns v, w . For a tangent vector (u, w) in total-space coordinates (z, v) , differentiation of $G_h(v) = v^\dagger H_h v$ gives

$$\text{Levi}(G_h)(u, w) = w^\dagger H_h w + 2\text{Re}(w^\dagger (H_h)_u v) + v^\dagger (H_h)_{u\bar{u}} v, \quad (2.7)$$

where $(H_h)_u := \sum_i u_i \partial_i H_h$ and $(H_h)_{u\bar{u}} := \sum_{i,j} u_i \bar{u}_j \partial_i \partial_{\bar{j}} H_h$. Since X is projective and hence compact, choose finitely many coordinate neighborhoods $\tilde{U}_\nu$, each carrying a holomorphic frame of F , and open sets $U_\nu \Subset \tilde{U}_\nu$ that cover X . If $H_h^{(\nu)}$ is the coefficient matrix in the frame on $\tilde{U}_\nu$, use the norm

$$\|h\|_{C^2} := \max_{\nu, a, b, |\alpha| \leq 2} \sup_{x \in \tilde{U}_\nu} |\partial^\alpha (H_h^{(\nu)})_{ab}(x)|,$$

where ∂^α denotes differentiation in the underlying real coordinates. For a continuous Hermitian form Φ on W , use

$$\|\Phi\|_{C^0} := \sup_{(v, \xi) \in \mathcal{K}} |\Phi_v(\xi, \xi)|,$$

the supremum of its operator norm with respect to b . Writing $\text{Herm } F$ and $\text{Herm } W$ for the real vector bundles of Hermitian forms on F and W , respectively, (2.7) gives a continuous real-linear operator

$$\mathcal{D} : C^2(X, \text{Herm } F) \longrightarrow C^0(S_k F, \text{Herm } W), \quad \|\mathcal{D}h\|_{C^0} \leq C \|h\|_{C^2}.$$

The constant C depends only on the fixed metrics, coordinate cover, and frames. For scalar functions on $\mathcal{K}$, write $\|f\|_\infty := \sup_{q \in \mathcal{K}} |f(q)|$. Since $\|f_h\|_\infty = \|\mathcal{D}h\|_{C^0}$, C^2 approximation of h gives uniform approximation of f_h on $\mathcal{K}$.

The C^2 -density follows from the standard approximation of Hermitian metrics by Fubini–Study metrics, stated in Hashimoto–Keller [11, Section 2.3, p. 11, following (2.9)]. Applied to the dual h^* of a smooth positive-definite metric h on F , this gives Fubini–Study metrics g_m on E with $g_m \rightarrow h^*$ in C^2 . Their duals g_m^* belong to $\mathcal{H}_m$: they are induced by positive-definite Hermitian forms on V_m^* through the dual evaluation map, with the twist removed by a^m . Consequently $g_m^* \rightarrow h$ in C^2 . Every smooth Hermitian form u is a difference $(u + Ck) - Ck$ of positive-definite metrics for a sufficiently large constant C . Approximating both at the same levels and subtracting, then using smooth approximation of C^2 forms, shows that $\bigcup_{m \geq m_0} \mathcal{H}_m$ is dense in the space of C^2 Hermitian forms on F . If $\dim X = 0$, this is immediate since every Hermitian form belongs to each $\mathcal{H}_m$.

Proof of Theorem 1.1. Proposition 2.1 proves the equivalence of (ii) and (iii). If (ii) holds, then $f_{h_{m,B}}(q) = \text{tr}(B\Psi_m(q)) > 0$ for every $q \in \mathcal{K}$. Normalizing the fibre point and tangent vector shows that $G_{h_{m,B}}$ is strictly plurisubharmonic on $F \setminus 0$. Equation (2.1) gives $h_{m,B} > 0$. Since $G_{h_{m,B}}$ is strictly plurisubharmonic on $F \setminus 0$, Drinovec Drnovšek–Forstnerič [6, Proposition 6.2(ii)] implies that $h_{m,B}$ is Griffiths negative. Its dual metric on E is Griffiths positive, proving (ii) $\Rightarrow$ (i).

Conversely, let h be the dual of a Griffiths-positive metric. Since G_h is strictly plurisubharmonic off the zero section, f_h has a positive minimum δ on the compact space $\mathcal{K}$. The

C^2 -density established above, together with the continuity following (2.7), gives $h_m \in \mathcal{H}_m$ for some m such that

$$\|f_{h_m} - f_h\|_\infty < \delta/2.$$

Thus $f_{h_m} > 0$, proving (i) $\Rightarrow$ (ii).

For the dual assertion, we use the representation of positive functionals by matrix-valued measures. The metric b identifies Hermitian forms on W with b -self-adjoint endomorphisms. Their real vector bundle is denoted by $\text{Herm}_b(W)$, and its identity section I_W corresponds to b . A positive real-linear functional T on $C^0(S_k F, \text{Herm}_b(W))$ is automatically continuous, since

$$|\langle T, \Phi \rangle| \leq \|\Phi\|_{C^0} \langle T, I_W \rangle. \quad (2.8)$$

Here Φ is a continuous Hermitian form on W , with the C^0 norm specified above. The Riesz representation theorem in local unitary frames identifies these functionals with positive matrix-valued Radon measures. In particular, $T \neq 0$ implies $\langle T, I_W \rangle > 0$, and $\Phi \geq \delta I_W$ with $\delta > 0$ implies $\langle T, \Phi \rangle > 0$ for every such nonzero T .

If (i) holds, the dual metric h is Griffiths negative, so $\mathcal{D}h > 0$. Compactness then gives $\mathcal{D}h \geq \delta I_W$ for some $\delta > 0$. For a nonzero positive matrix-valued measure T , (2.8) therefore gives

$$\langle T, \mathcal{D}h \rangle \geq \delta \langle T, I_W \rangle > 0.$$

Such a T cannot satisfy $\mathcal{D}^*T = 0$, proving (i) $\Rightarrow$ (iv).

Suppose now that every level fails. By Proposition 2.1, $0 \in \text{conv } \Psi_m(\mathcal{K})$ for every $m \geq m_0$. At each level choose points q_j and weights λ_j satisfying (2.6), and let $\mu_m := \sum_{j=1}^s \lambda_j \delta_{q_j}$, where δ_{q_j} is the unit point mass at q_j . This is a probability measure on $\mathcal{K}$. For every $h = h_{m,B} \in \mathcal{H}_m = \{h_{m,B} : B \in \text{Herm}(N_m)\}$, the first-jet identity gives

$$\int_{\mathcal{K}} f_h d\mu_m = \sum_{j=1}^s \lambda_j \text{tr}(B \Psi_m(q_j)) = \text{tr} \left(B \sum_{j=1}^s \lambda_j \Psi_m(q_j) \right) = 0.$$

Since $\mathcal{K}$ is compact and metrizable, a subsequence μ_{m_j} converges weakly to a probability measure μ . For a fixed level $m' \geq m_0$ and $h \in \mathcal{H}_{m'}$, the inclusion $\mathcal{H}_{m'} \subset \mathcal{H}_m$ for $m \geq m'$ gives

$$\int_{\mathcal{K}} f_h d\mu_m = 0 \quad (m \geq m').$$

Passing to the limit shows that μ annihilates every finite level. The C^2 -density and the C^2 -to- C^0 continuity of $\mathcal{D}$ extend this identity to all smooth Hermitian forms.

The formula

$$\langle T, \Phi \rangle := \int_{\mathcal{K}} \Phi_v(\xi, \xi) d\mu(v, \xi)$$

gives a positive matrix-valued Radon measure on $S_k F$. Since $b(\xi, \xi) = 1$, its trace mass is $\langle T, I_W \rangle = 1$, and

$$\langle \mathcal{D}^*T, h \rangle = \langle T, \mathcal{D}h \rangle = \int_{\mathcal{K}} f_h d\mu = 0.$$

Thus any sequence of the chosen atomic measures has a subsequence yielding a nonzero T with $\mathcal{D}^*T = 0$. If (i) fails, every level fails because (ii) implies (i). The resulting nonzero T contradicts (iv). This proves (iv) $\Rightarrow$ (i) and completes the equivalence of (i)–(iv). $\square$

3. AN APPLICATION ON AN ABELIAN SURFACE

3.1. Construction and failure at level zero. The obstruction at level zero is detected by the differential of the complete evaluation map. For a globally generated bundle E , take $m_0 = 0$ and write $V := V_0 = H^0(X, E)$, with $F = E^*$ as before. The evaluation morphism is the holomorphic map

$$\hat{\Phi} : F \longrightarrow V^*, \quad \hat{\Phi}(x, v)(s) := v(s(x)).$$

Equip V^* with a constant Hermitian inner product and use an orthonormal basis. The first-jet identity (1.2) at $m = 0$ gives

$$\text{Levi}(G_{0,B})(\xi, \xi) = d\widehat{\Phi}(\xi)^\dagger B d\widehat{\Phi}(\xi).$$

Thus the complete level-zero test succeeds precisely when $\widehat{\Phi}$ is an immersion on $F \setminus 0$: injectivity of its differential gives strict positivity for $B = I$, whereas a nonzero kernel vector makes the displayed expression vanish for every Hermitian B .

The rows and columns of the Steiner matrix T below are indexed starting at zero. On $\mathbb{P}^2$ with coordinates $[x : y : z]$, consider

$$\mathsf{T}(x, y, z) := \begin{pmatrix} x & 0 & 0 & z & 0 \\ 0 & x & 0 & 0 & z \\ y & 0 & x & 0 & 0 \\ 0 & y & 0 & x & 0 \\ -z & 0 & y & 0 & x \\ 0 & z & 0 & y & 0 \\ 0 & 0 & z & 0 & y \end{pmatrix}. \quad (3.1)$$

The cokernel S occurs in the sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-1)^5 \xrightarrow{\mathsf{T}} \mathcal{O}_{\mathbb{P}^2}^7 \longrightarrow S \longrightarrow 0. \quad (3.2)$$

This is the type of Steiner quotient used in Du–Xie [7]; we prove the properties needed below directly from the displayed matrix.

An elementary row calculation in (3.1) gives $\text{rank } \mathsf{T}(x, y, z) = 5$ at every point of $\mathbb{P}^2$, so S is locally free of rank two.

For $v = (v_0, \dots, v_6)^t \in \mathbb{C}^7$, set

$$M_v := \begin{pmatrix} v_0 & v_2 & -v_4 \\ v_1 & v_3 & v_5 \\ v_2 & v_4 & v_6 \\ v_3 & v_5 & v_0 \\ v_4 & v_6 & v_1 \end{pmatrix}.$$

The identity

$$\mathsf{T}(x, y, z)^t v = M_v(x, y, z)^t$$

is immediate entry by entry.

For a nonzero homogeneous coordinate column $Z := (x, y, z)^t \in \mathbb{C}^3$, the dual fibres are

$$S_{[Z]}^* = \ker \mathsf{T}(Z)^t = \{v \in \mathbb{C}^7 : M_v Z = 0\}.$$

Let h_S be the metric on S^* induced by the standard metric on $\mathbb{C}^7$, and $g_S := h_S^*$ its dual. The holomorphic map $S^* \setminus 0 \rightarrow \mathbb{C}^7$, $([Z], v) \mapsto v$, is an immersion: a tangent vector in its differential kernel satisfies $\dot{v} = 0$ and $M_v \dot{Z} = 0$, while $\text{rank } M_v \geq 2$ for $v \neq 0$ and $M_v Z = 0$ give $\ker M_v = \mathbb{C}Z$, so the base tangent vector also vanishes. Thus $G_{h_S} = |v|^2$ is strictly plurisubharmonic, and the curvature–Levi correspondence used in Section 2 shows that g_S is Griffiths positive.

For a Hermitian metric h with local coefficient matrix H_h , we use the convention in which the local matrix of its Chern curvature Θ_h is $i \sum R_{i\bar{j}}^h dz_i \wedge d\bar{z}_j$, where $R_{i\bar{j}}^h := -\partial_{\bar{j}}(H_h^{-1} \partial_i H_h)$, as in Kobayashi [12, Chapter I, Section 4]. Write $\omega_{\text{FS}} = i\partial\bar{\partial} \log |Z|^2$. Compactness of $\mathbb{P}^2$ gives a constant $c_S > 0$ such that, in the Griffiths sense,

$$\Theta_{g_S} \geq c_S \omega_{\text{FS}} \otimes I_S. \quad (3.3)$$

We now construct the map $f : X \rightarrow \mathbb{P}^2$ and the bundle $E = f^*S$ in Example 1.2, and verify the failure at level zero. Let C be the smooth projective completion of $y^2 = x^3 - x$, with point ∞ . The meromorphic functions x, y have their only poles at ∞ , of orders 2, 3.

The curve has genus one. By Riemann–Roch and Serre duality [9, Chapter IV, Theorem 1.3 and Example 1.3.6], a positive-degree line bundle L satisfies $h^0(L) = \deg L$ and $H^1(L) = 0$.

On the two factors of $X = C \times C$, choose the bases

$$(a_0, a_1, a_2, a_3) := (1, x_1, y_1, x_1^2), \quad (b_0, b_1, b_2, b_3, b_4) := (1, x_2, y_2, x_2^2, x_2 y_2).$$

Their distinct pole orders show linear independence, and the dimension formula shows that they are complete bases for $\mathcal{O}_C(4[\infty])$ and $\mathcal{O}_C(5[\infty])$. The twenty products

$$t_{ab} := a_a b_b, \quad 0 \leq a \leq 3, \quad 0 \leq b \leq 4,$$

form a basis of $H^0(X, H)$ for

$$H = \mathcal{O}_C(4[\infty]) \boxtimes \mathcal{O}_C(5[\infty]).$$

Both factors are very ample by the degree criterion for curves [9, Chapter IV, Corollary 3.2(b)], since their degrees are at least $2g(C) + 1 = 3$. Their exterior product is very ample by the product embeddings followed by the Segre embedding. In particular, the complete system of the t_{ab} has positive Fubini–Study curvature.

Consider the sections and the associated map

$$f_0 := t_{00}, \quad f_1 := t_{34}, \quad f_2 := t_{30} + t_{04}, \quad f := [f_0 : f_1 : f_2] : X \longrightarrow \mathbb{P}^2. \quad (3.4)$$

The pairs $(1, x^2)$ and $(1, xy)$ have no common zero as sections of $\mathcal{O}_C(4[\infty])$ and $\mathcal{O}_C(5[\infty])$, respectively: the second section in each pair is nonzero at infinity by its pole order. They give morphisms $\pi_4 := [1 : x^2]$ and $\pi_5 := [1 : xy]$ to $\mathbb{P}^1$. The three sections in (3.4) arise by composing $\pi_4 \times \pi_5$ with

$$([u_0 : u_1], [v_0 : v_1]) \longmapsto [u_0 v_0 : u_1 v_1 : u_1 v_0 + u_0 v_1].$$

The three displayed coordinates have no common zero on $\mathbb{P}^1 \times \mathbb{P}^1$, so f is a morphism and $f^* \mathcal{O}_{\mathbb{P}^2}(1) = H$.

Since the quotient in (3.2) is locally free, the pullback $E = f^* S$ has the exact resolution

$$0 \longrightarrow (H^{-1})^5 \longrightarrow \mathcal{O}_X^7 \longrightarrow E \longrightarrow 0. \quad (3.5)$$

On an elliptic curve a negative-degree line bundle has no sections. The Künneth formula applied to the two negative-degree factors of H^{-1} therefore gives

$$H^0(X, H^{-1}) = H^1(X, H^{-1}) = 0.$$

The long exact sequence of (3.5) now gives $H^0(X, E) = \mathbb{C}^7$. Denote the images of the seven standard basis sections of $\mathcal{O}_X^7$ by $\sigma_0, \dots, \sigma_6$. They are the pullbacks of the seven quotient sections of S .

For the differential kernel in Example 1.2(i), take

$$P := ((0, 0), (2, \sqrt{6})) \in X.$$

On the first elliptic curve, y_1 is a local parameter at $(0, 0)$, and differentiating $y_1^2 = x_1^3 - x_1$ gives $dx_1 = 0$ there. Both nonconstant coordinates of f depend on this factor through x_1^2 , so

$$df_P(u) = 0, \quad u := \partial/\partial y_1 \neq 0.$$

Choose any unit vector $v \in S_{f(P)}^*$.

In the inclusion $E^* \subset X \times \mathbb{C}^7$, the vector $\xi := (u, \dot{v} = 0)$ is tangent to the total space: the derivative of its defining equation $\mathsf{T}(f(x))^t v = 0$ vanishes because $df_P(u) = 0$. Since these seven quotient sections form a basis of $H^0(X, E)$, the complete level-zero evaluation is precisely $(x, v) \mapsto v$. Its derivative kills ξ . With background $k = f^* h_S$ and the tangent metric b , the test space $\mathcal{K}$ is the unit sphere bundle from (1.1). Thus $q_0 := (v, \xi/\sqrt{b(\xi, \xi)}) \in \mathcal{K}$ satisfies

$$\Psi_0(q_0) = 0. \quad (3.6)$$

This excludes *every* Hermitian matrix at level zero and gives an atomic obstruction supported at one point. This proves Example 1.2(i). We next construct a first-level form and its coefficient matrix to prove (ii).

3.2. Success at level one. For the sections f_0, f_1, f_2 in (3.4) and the product basis (t_{ab}) of $H^0(X, H)$, use their coefficients in a common local frame e_H of H to write

$$r := |f_0|^2 + |f_1|^2 + |f_2|^2, \quad s := \sum_{a=0}^3 \sum_{b=0}^4 |t_{ab}|^2.$$

Neither expression vanishes, and s/r is a positive smooth function on X . The forms

$$\omega_r := i\partial\bar{\partial}\log r = f^*\omega_{\text{FS}} \geq 0, \quad \omega_s := i\partial\bar{\partial}\log s > 0$$

are global; the strict positivity of ω_s follows from the very ampleness of H . For $M := \max_X(s/r)$ and the constant c_S in (3.3), choose

$$0 < \varepsilon M < \min\{c_S, \frac{1}{2}\}.$$

Let a_ε be the metric on H specified by $|e_H|_{a_\varepsilon}^2 = (r + \varepsilon s)^{-1}$, and set

$$h_1 := \frac{r + 2\varepsilon s}{r + \varepsilon s} f^* h_S, \quad g_1 := h_1^*.$$

Here h_1 is a metric on $F = E^* = f^* S^*$, and g_1 is its dual on E .

Index the product basis of $H^0(X, H)$ by $j := 5a + b$, so $0 \leq j \leq 19$, and write $t_j := t_{ab}$. For the standard coordinate columns e_j of $\mathbb{R}^{20}$, the matrix

$$R := e_0 e_0^t + e_{19} e_{19}^t + (e_{15} + e_4)(e_{15} + e_4)^t$$

satisfies $r = t^\dagger R t$ and $s = t^\dagger t$, where $t := (t_0, \dots, t_{19})^t$. Since $R + \varepsilon I_{20} > 0$, the section inner product with Gram matrix $(R + \varepsilon I_{20})^{-1}$ makes a_ε a Fubini–Study metric as in (2.5). From now on the spaces $\mathcal{H}_m$ and maps Ψ_m are taken with $A = H$ and $a = a_\varepsilon$, so $e^\phi = r + \varepsilon s$ in (2.3).

To check the curvature of g_1 , write $\eta := \log(s/r)$ and $\tau := \varepsilon e^\eta$. The function $\chi(\eta) := \log(1 + 2\varepsilon e^\eta) - \log(1 + \varepsilon e^\eta)$ has derivatives

$$\alpha(\tau) := \chi'(\eta) = \frac{\tau}{(1 + \tau)(1 + 2\tau)}, \quad \beta(\tau) := \chi''(\eta) = \frac{\tau(1 - 2\tau^2)}{(1 + \tau)^2(1 + 2\tau)^2}.$$

Our choice of ε gives $0 < \alpha(\tau) < c_S$ and $\beta(\tau) > 0$. Since $g_1 = e^{-\chi(\eta)} f^* g_S$, the Chern curvature formula and (3.3) yield

$$\begin{aligned} \Theta_{g_1} &= f^* \Theta_{g_S} + [\alpha(\tau)(\omega_s - \omega_r) + \beta(\tau)i\partial\eta \wedge \bar{\partial}\eta] \otimes I_E \\ &\geq (c_S - \alpha(\tau))\omega_r \otimes I_E + \alpha(\tau)\omega_s \otimes I_E > 0. \end{aligned}$$

Thus h_1 is Griffiths negative, including in the directions killed by df . We now express it by a finite-level coefficient matrix.

At level one choose a basis $s_1^{(1)}, \dots, s_{N_1}^{(1)}$ of $V_1 = H^0(X, E \otimes H)$, where $N_1 = \dim V_1$, and write the product sections in this basis:

$$\sigma_i t_j = \sum_{\alpha=1}^{N_1} c_{ij,\alpha} s_\alpha^{(1)}, \quad 0 \leq i \leq 6, \quad 0 \leq j \leq 19.$$

Let $\mathbf{C} := (c_{ij,\alpha})$, with rows ordered first by i and then by j . For $v \in F_x$, the evaluation column $p_1(x, v)$ records the coefficients of $v(s_\alpha^{(1)}(x))$ in the chosen local frame of H . The column of product evaluations is therefore $\mathbf{C}p_1$. Consider

$$B_1 := \mathbf{C}^\dagger [I_7 \otimes (R + 2\varepsilon I_{20})] \mathbf{C}. \quad (3.7)$$

Since h_S is induced by the standard metric on $\mathbb{C}^7$, one has $G_{f^* h_S}(v) = \sum_{i=0}^6 |v(\sigma_i)|^2$. Thus B_1 is Hermitian and

$$p_1^\dagger B_1 p_1 = \sum_{i=0}^6 |v(\sigma_i)|^2 t^\dagger (R + 2\varepsilon I_{20}) t = (r + 2\varepsilon s) G_{f^* h_S}(v).$$

Since $e^\phi = r + \varepsilon s$ for a_ε , division by this denominator gives $G_{1,B_1} = G_{h_1}$ in the notation of (2.3). This also shows that $h_1 \in \mathcal{H}_1$. Its Griffiths negativity proved above implies that G_{h_1} is strictly plurisubharmonic off the zero section. The first-jet identity (1.2) therefore gives

$$\mathrm{tr}(B_1 \Psi_1(q)) = f_{h_1}(q) > 0 \quad (q \in \mathcal{K}),$$

which proves Example 1.2(ii). Theorem 1.1 applied to this matrix yields Griffiths positivity of E . At level zero, the evaluation map $\widehat{\Phi} : E^* \setminus 0 \rightarrow H^0(X, E)^*$, given by $\widehat{\Phi}(x, v)(s) = v(s(x))$, has a nonzero differential kernel; this gives (3.6) and rules out every Hermitian matrix B at that level.

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